Extensive Form Games

An *extensive form game* is a collection $\Gamma = (N, A, H, \iota, \mathbb{I}, \{U_i\}_{i \in N})$.

1. $N = \{1, ..., n\}$, a finite set of *players.*
2. A, a nonempty set of *actions.*
3. H, a set of histories (sometimes called nodes), with
   (a) $h_0 \in H$, where h_0 is the *empty history.*
   (b) $h \in H - \{h_0\}$ *only if* $h = h_0, a_1, ..., a_k$ where $a_\ell \in A$ for $\ell = 1, ..., k$, and k is either a positive integer or $+\infty$.
   (c) If $h' \in H$ and $h' = h, a_1, ..., a_k$, then $h \in H$.

Let

- $T = \{h \in H \,|\, h$ is an infinite sequence; or h is finite and $(h, a) \notin H$ for any $a \in A\}$. T is the set of *terminal histories* (or *outcomes*); and
- $\mathcal{A}(h) = \{a \in A \,|\, (h, a) \in H\}$ is the set of actions available at $h \in H$.

4. A function $\iota : H \backslash T \to N$ giving the player with the move at each non-terminal history.
5. A partition $\mathbb{I}$ of $H \backslash T$ such that $\{h_0\} \in \mathbb{I}$, and for any two members h, h' of the same partition element (i) $\mathcal{A}(h) = \mathcal{A}(h')$; (ii) $\iota(h) = \iota(h')$. Each element of $\mathbb{I}$ is an *information set.* For any $h \in H \backslash T$, let $I(h)$ denote the set of histories in H that are in the same information set as h.
6. $U_i : T \to \mathbb{R}$, a utility function for each player $i \in N$, representing i's preferences over the set T of terminal histories. (Often preferences are defined over the set ΔT of lotteries over terminal histories and U_i is taken to be a vN-M utility.)

The Strategic Form Representation of Γ. Denote by $\mathbb{I}_j$ the set of player $j's$ information sets: $\mathbb{I}_j = \{I \in \mathbb{I} \,|\, \iota(I) = \{j\}\}$. A *strategy* for player $j \in N$ in game Γ is a function $s_j : \mathbb{I}_j \to A$ with $s_j(I) \in \mathcal{A}(h)$ for all $h \in I$ and all $I \in \mathbb{I}_j$. In words: a strategy for player j is a function from that j's information sets to feasible actions.[1] A player's strategy set $S_i = \Pi_{\{h \in H | \iota(h) = i\}} \mathcal{A}(h)$ is the set of all feasible strategies. A strategy profile $s \in S = S_1 \times \cdots S_n$ generates a terminal history $t \in T$. Player i's preferences over $S = S_1 \times \cdots S_n$ are represented by $u_i(s) = U_i(t)$, where $t \in T$ is the terminal history generated by $s \in S$. An *outcome* of a game is an element of T (often written with the initial history h_0 omitted).

Subgame Perfection. A Nash equilibrium is *subgame perfect* if it generates a Nash equilibrium on each subgame. I write that $h' \in H$ or *follows* (or succeeds) $h \in H$ ($h' \mathcal{F} h$) if $h' = ha'_1, ..., a'_k$. A *subgame* of Γ starts at a history $h \in H$ such that $\{h\} \in \mathbb{I}$, and for any $h' \mathcal{F} h$, if $h'' \in I(h')$, then $h'' \mathcal{F} h$. Informally: a subgame starts from a singleton information set such that no subsequent information sets are 'sliced.'

Note well: *Outcomes* and *strategy profiles* are not the same. For example in game Γ_2 from class, the unique subgame perfect equilibrium *outcome* is $h_0 Lr$ (often shortened to Lr). Since 'r' is not a strategy for player 2, 'Lr' is not a strategy profile (hence not an Nash equilibrium).

[1] Alternatively, a strategy for j could be defined as a function s_j from $\iota^{-1}(j)$ into A such that s_j is constant on each $I \in \mathbb{I}_j$ and $s_j(h) \in \mathcal{A}(h)$ for each $h \in \iota^{-1}(j)$.